MIL-T-8409A(USAF)
29 September 1969
SUPERSEDING
MIL-T-8409(USAF)
28 July 1958

MILITARY SPECIFICATION

THERMOMETER, BIMETAL, DIRECT READING, CABIN, AIR, -50°C TO +70°C, TYPE C-17

1. SCOPE

1.1 This specification covers the 1-7/8 inch round dial, bimetal type, direct reading thermometer, designated type C-17.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings Engineering And Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

Military

MIL-STD-192	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards, Order Of Precedence For The Selection Of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

MIL-STD-1186	**Cushioning**, Anchoring, Bracing, Blocking, **And** Waterproofing; With Appropriate Test **Methods**
MS33558	Numerals And Letters, Aircraft Instrument Dial, Standard Form Of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules And Regulations

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago IL 60606.)

3. REQUIREMENTS

* 3.1 Preproduction testing. This specification makes provisions for preproduction testing.

* 3.2 Specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

* 3.2.1 Standard parts. AN or MIL Standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used provided they have suitable properties and are replaceable by the AN or MIL Standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

* 3.3 Materials.

* 3.3.1 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

* 3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the thermometer, except where magnetic materials are essential.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

* 3.3.3 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the Procuring Agency.

* 3.3.4 Protective treatment. When materials are used in the construction of the thermometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

* 3.4 Design. The thermometer shall be designed to indicate the temperature of the surrounding air and shall conform to Figure 1. The complete thermometer shall consist of the thermometer and mounting bracket as shown on Figure 1.

* 3.5 Construction. The thermometer shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

* 3.5.1 Pointer. The pointer shall be sufficiently rigid to prevent oscillation or flexing under vibration. Pointer stops shall not be used.

* 3.5.2 Dial. The dial shall be securely fastened to the case or frame of the mechanism in such a manner that it will not become loose or turn when the thermometer is vibrated.

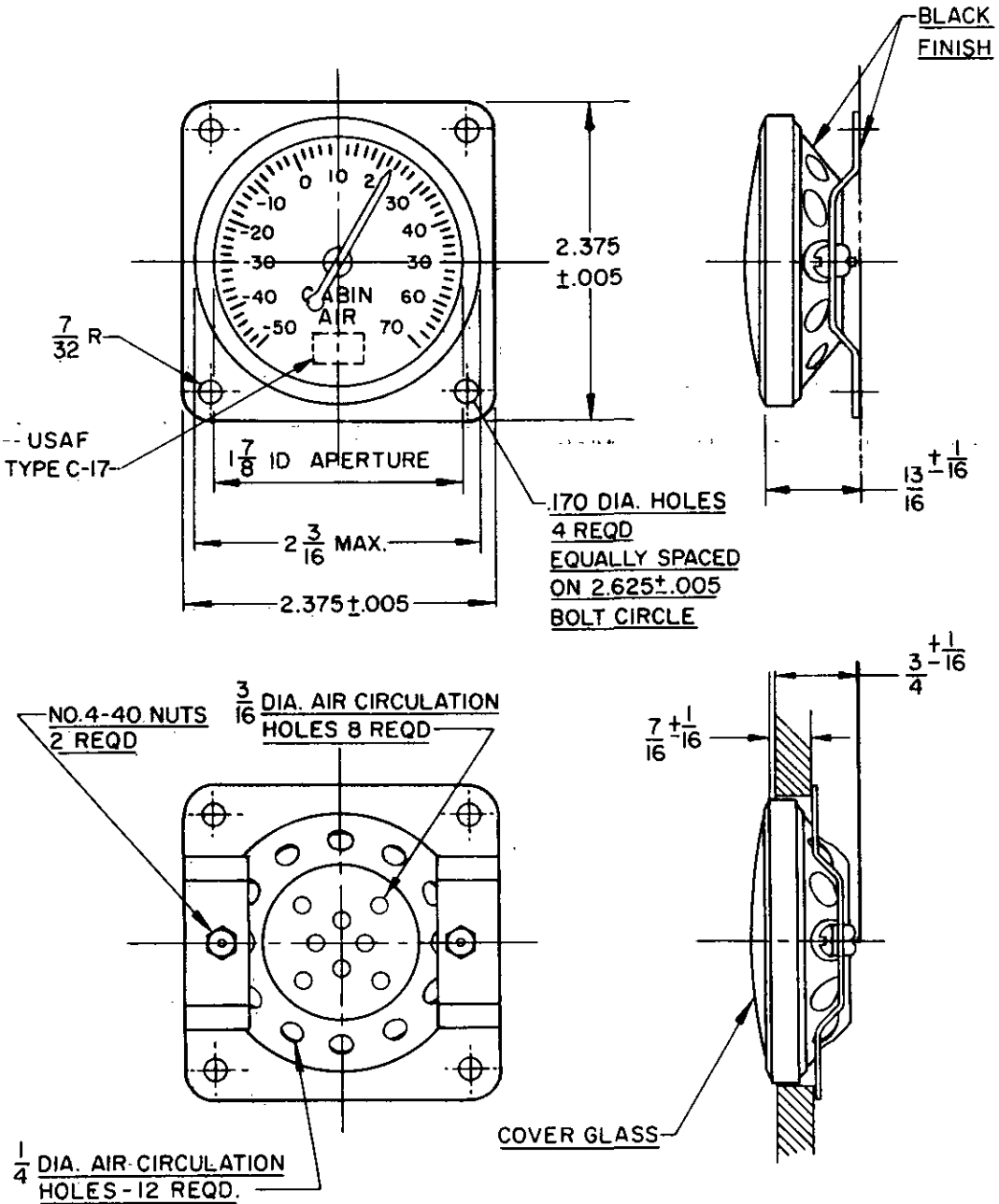
* 3.5.2.1 Markings. All markings shall be durable to withstand usage encountered in service. The pointer, all graduations, numerals, and other specified markings on the dial shall be visible from any point within a frustum of a cone the side of which makes an angle of 30 degrees with a perpendicular to the dial and the small diameter of which is the aperture of the thermometer case. Numerals and letters on the dial shall conform to MS33558.

* 3.5.2.2 Numerals. Numerals shall distinctly indicate the graduation to which each applies. When several numerals are used in one group, the space between the numerals shall be not less than 1/64 inch.

* 3.5.2.3 Fluorescent-luminescent material. The shaded portion of the pointer and the following markings shown on Figure 1 shall be finished in fluorescent-luminescent material conforming to MIL-L-25142. The dimensions of the markings shall be as follows:

	Height or Length, Inch ±1/64	Width of Line or Graduation, Inch ±0.005
Numerals	1/8	0.020
10-degree graduations	1/8	0.025
Pointer	---	0.062
2-degree graduations	1/16	0.015
Lettering "°C CABIN AIR"	1/8	0.020

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.



NOTE: MOUINTING BRACKET MAY BE TURNED FOR MOUNTING ON A WALL OR ON A PANEL.

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED

TOLERANCE ON FRACTIONS: $\pm \frac{1}{64}$

FIGURE 1. Thermometer and Mounting Bracket

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

* 3.5.2.4 Durable dull black. The following markings shall be permanently and legibly marked on the dial. These markings and all other markings not otherwise specified shall be finished in durable dull black. The height of the markings shall be as follows:

	Height, Inch
USAF	3/64
TYPE C-17	3/64

* 3.5.3 Cover glass. The cover glass shall be clear and free from flaws.

* 3.5.4 Spacing - pointer, dial, cover glass. The distance between the dial and cover glass shall be the minimum practicable. The distance between the pointer and dial shall also be the minimum practicable to minimize parallax.

* 3.6 Performance. The thermometer shall be capable of meeting the test requirements specified in Section 4.

* 3.7 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

* 3.8 Weight. The weight of the complete thermometer shall not exceed 0.20 pound.

* 3.9 Screw threads. The threads of all machine screws greater than 0.060 inch in diameter shall conform to MIL-S-7742.

* 3.10 Finishes and protective coating.

* 3.10.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial and small holes need not be anodized.

* 3.10.2 The background of the dial and the front of the thermometer case shall be finished in durable dull black.

* 3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 with the following information.

Specification MIL-T-8409
Order No. _____
Mfr's. Part No._____
Mfr's. Name or Trademark

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

* 3.12 Workmanship.

* 3.12.1 General. The thermometer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the thermometer shall be classified as follows:

a.	Preproduction testing	See 4.4
b.	Quality conformance inspection	See 4.5

* 4.3 Test conditions.

* 4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches mercury and at room temperature (approximately +25° Centigrade (C)). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

* 4.3.2 The thermometer shall be lightly vibrated or tapped before a test reading is taken.

* 4.3.3 Vibration stand. Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 300 and 3000 cycles per minute and shall subject the thermometer to vibration such that a point on the thermometer case will describe in a plane inclined 45 degrees to the horizontal plane, a circle of not less than 0.018 nor more than 0.020 inch in diameter.

* 4.3.4 The drift, durability, and response time tests shall be conducted in air moving at a velocity of 20 ±5 feet per second.

* 4.4 Preproduction testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

* 4.4.1 Preproduction test samples. Preproduction test samples shall consist of 3 thermometers. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required.

* 4.4.2 The preproduction test shall consist of all the tests specified herein.

* 4.5 Quality conformance inspection.

* 4.5.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The tests shall be accomplished on articles to be supplied on the contract or order. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

* 4.5.2 Individual tests. Each thermometer shall be subjected to the following tests.

* 4.6 Test methods.

* 4.6.1 Examination of product. Each thermometer shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

* 4.6.2 Scale error. The bimetal sub-assembly shall be immersed in an agitated liquid bath and tested at the points of the scale indicated in Table I before it is assembled in the thermometer case. The sub-assembly shall be tapped before readings are taken and the friction shall not exceed 1°C at any test point.

TABLE I

True Temperature °C	Tolerance
-30	±2°
0	±2°
+30	±2°

* 4.6.3 The bimetal sub-assembly shall then be installed in the thermometer case and the pointer set at the existing room temperature. Care shall be exercised that the bimetal has remained at the existing room temperature long enough to stabilize. The room temperature shall be constant. The thermometer shall then be subjected to air temperatures of -50 ±3°C and +70 ±3°C to check its movement over the scale range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

* 4.6.4 Position error. The reading of the thermometer at room temperature shall be observed in the normal operating position with the dial in the vertical plane. The thermometer shall then be rotated through angles of 90 degrees to right and left, and also tipped to a position with the horizontal. The change in indication when the thermometer is held in any of the above positions shall not exceed 1°C.

* 4.6.5 Sampling tests. Five thermometers which have been manufactured under essentially the same conditions shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests.

* 4.6.5.1 When sampling tests are specified on a number of thermometers that are selected as representative of a certain lot, and one or more of this number fail to meet the specified tests, acceptance of all thermometers shall be withheld until the extent and cause of failure are determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure, but the final acceptance of the thermometer is contingent upon the inspector's decision regarding the over-all conformance of the thermometer to specification requirements. If investigation indicates that the defects may exist on thermometers previously accepted, full particulars concerning the defects found, including recommendations for correction, will be furnished to the Procuring Agency.

* 4.6.5.2 Vibration error. The thermometer shall be subjected to vibration error in accordance with Procedure IV of MIL-E-5272. The diameter of circular motion shall be 0.018 to 0.020 inch. The pointer oscillation shall not exceed 3°C.

* 4.6.5.3 Vibration failure. The thermometer shall be subjected to vibration failure in accordance with Procedure V of MIL-E-5272. After completion of the test, the thermometer shall be subjected to and shall meet the requirements specified in 4.6.2.

* 4.6.6 Special tests.

* 4.6.6.1 Three thermometers shall be selected at random from the first 15 thermometers representing the manufacturer's contract or order and subjected to all of the special tests.

* 4.6.6.2 Detail examination. The thermometer shall be critically examined to determine full compliance with regard to weight, physical dimensions, dial composition, markings, and similar detail requirements.

* 4.6.6.3 Drift. The thermometer shall be subjected to a temperature of +70 ±2°C for a period of 3 hours, after which the thermometer shall be subjected to a temperature of 0°C. The maximum change in scale error at this point, over that obtained in the tests specified in 4.6.2 shall not exceed 1/2°C. In addition, no failures of any kind shall occur.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

* 4.6.6.4 Durability. The thermometer shall be alternately subjected to agitated air at +30 ±2°C and -30 ±2°C 100 times, after which the scale error test of 4.6.2 shall be repeated. The change in indication at the conclusion of this test shall not be increased over indications obtained as a result of the scale error test specified in 4.6.2 by more than 1°C.

* 4.6.6.5 Response time (lag test). The thermometer shall be quickly transferred from a temperature of +50°C to a temperature of -30°C. The time required for the indicated temperature to fall from +50 to -20°C shall not exceed 120 seconds.

* 4.6.6.6 Humidity. The thermometer shall be subjected to a humidity test in accordance with Procedure I of MIL-E-5272.

* 4.6.6.7 Fungus resistance. The thermometer shall be subjected to a fungus resitance test in accordance with Procedure I of MIL-E-5272.

* 4.7 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

* 4.8 Inspection of preparation for delivery. The preservation, packaging, packing and marking for shipment and storage shall be examined and tested to determine compliance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

* 5.1.1 Level A. Each thermometer shall be preserved and packaged in accordance with MIL-P-116, Method III and placed in a snug fitting container conforming to Grade W6C, PPP-B-636.

* 5.1.2 Level C. Each thermometer shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it fulfills this requirement.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Thermometers preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and cube, shall comply to the applicable special requirements table of the specification. Interior cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix of PPP-B-636. Where practicable, the exterior shipping containers shall be of uniform shape and size and of minimum tare and cube.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

* 5.2.2 Level B. Same as Level A except that waterproofing is not required and domestic shipping containers may be used.

* 5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

* 5.3 Marking. The interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. The Type C-17 thermometer covered by this specification is a direct reading bimetal type and is intended for use in the aircraft to indicate the cabin or cockpit temperature.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Preproduction requirements (see 3.1).

c. Selection of applicable levels of preservation, packaging, and packing (see Section 5).

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

Review Activity:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 6685-F332

☆ U. S. GOVERNMENT PRINTING OFFICE: 1969-393-066/S-2202

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS

SAN ANTONIO AIR MATERIEL AREA
SERVICE ENGINEERING DIVISION (SANE)
KELLY AFB TEX 78241

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:23Z
Check the source to verify that this is the current version before use.